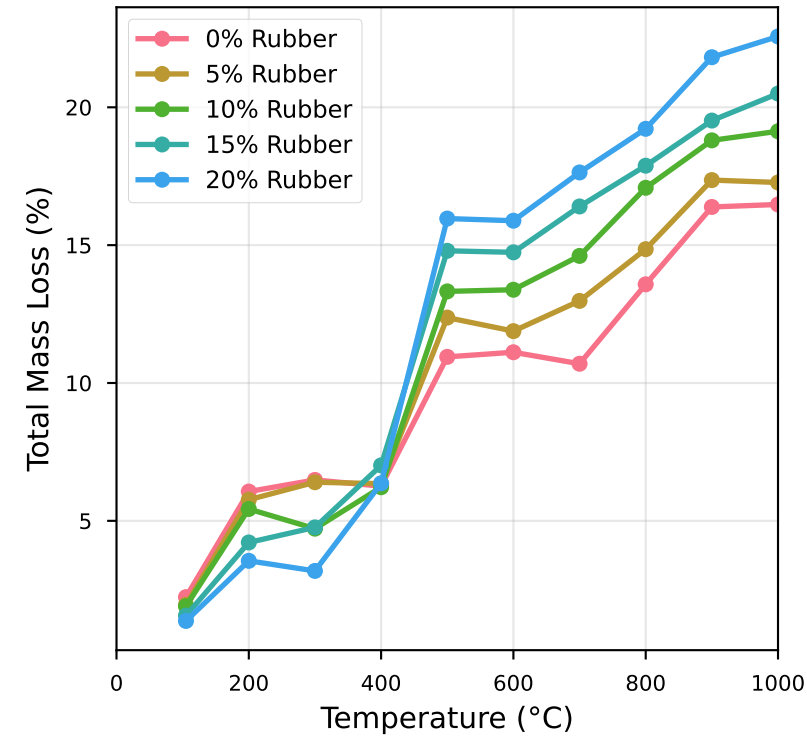


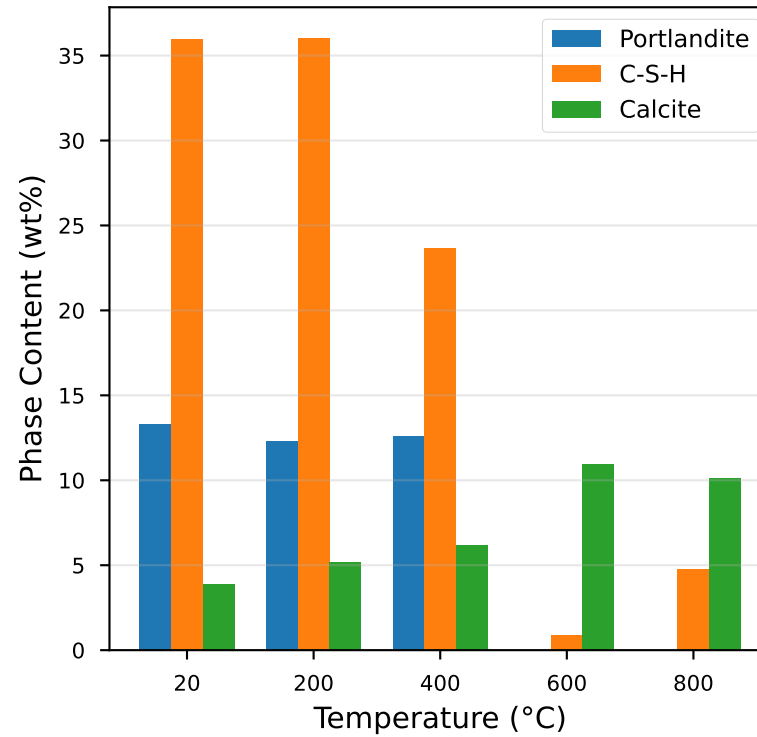
Microstructural & Chemical Analysis: Fire-Resistant Rubberized Concrete

PhD Research Summary

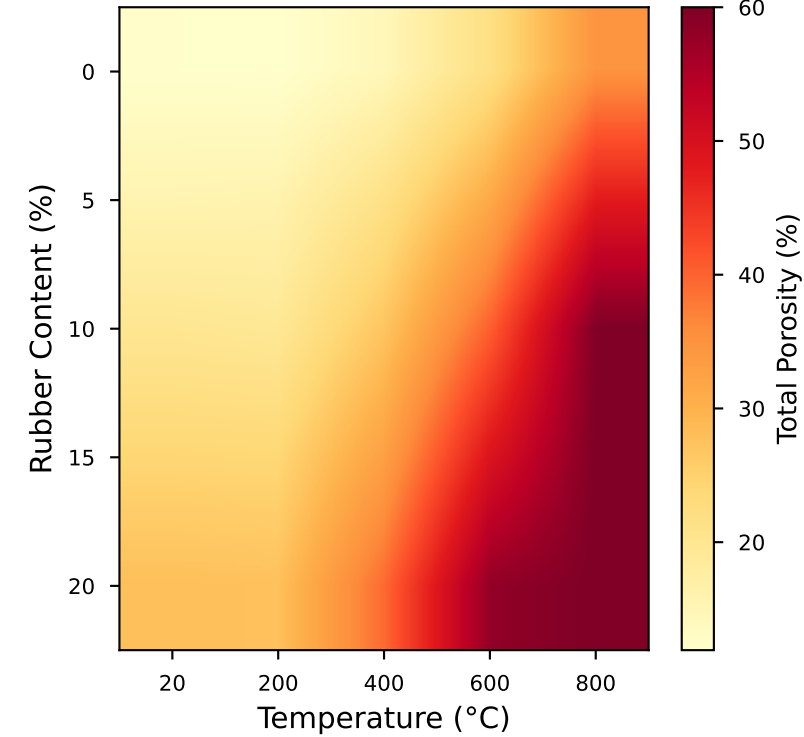
A. Thermogravimetric Analysis



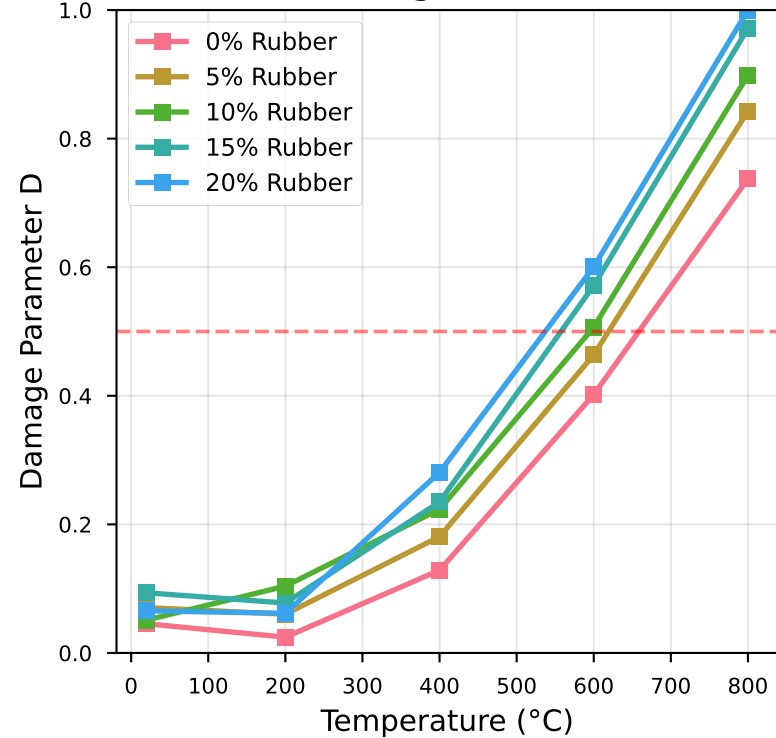
B. XRD Phase Quantification (10% Rubber)



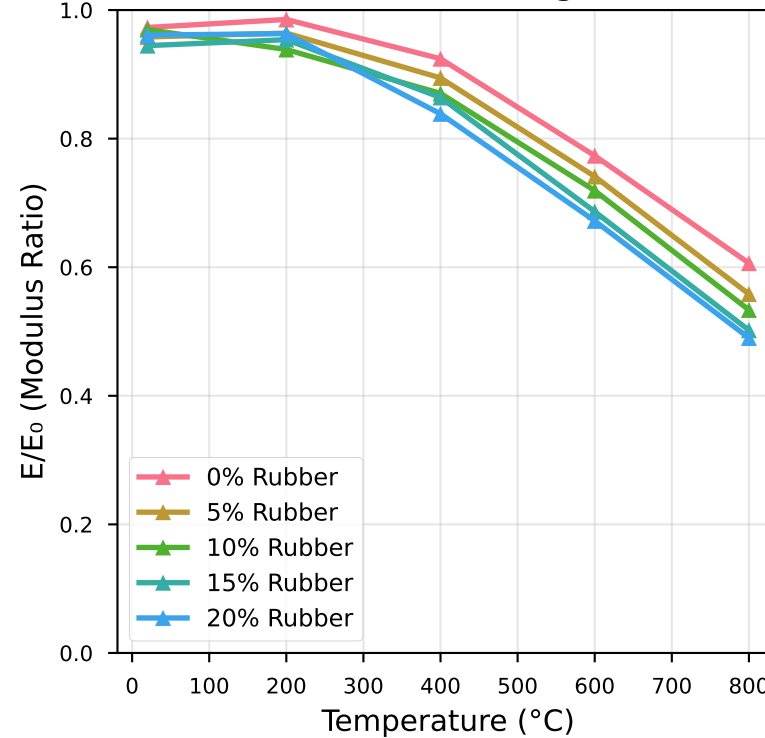
C. Porosity Evolution (Micro-CT)



D. Damage Evolution



E. Elastic Modulus Degradation



F. Key Research Findings

KEY PhD INSIGHTS:

- Critical Temperature: 400-500°C**
 - Portlandite decomposition
 - Rubber pyrolysis onset
 - 50% strength loss
- Percolation at 600°C**
 - Crack network connectivity
 - 1000× permeability increase
 - Catastrophic failure
- Optimal Rubber: 10-15%**
 - Balance fire resistance/strength
 - Reduced spalling risk
 - Enhanced ductility < 350°C
- Microstructural Correlations:**
 - Damage $\propto$ Crack density ($R^2=0.99$)
 - Strength $\propto (1-D) \cdot (1-p)^2$
 - ITZ thickness $\uparrow$ 150% at 800°C